An Enhanced IoT-Based Healthcare Monitoring System for Obese Adults

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Abstract

This paper presents an enhanced Internet-of-Things (IoT) healthcare monitoring prototype for obese adults, inspired by the work of Gupta et al. [1]. The prototype combines a DS18B20 digital ambient-temperature probe, a MAX30102 pulse-oximeter/heart-rate sensor, and a dual-core ESP32 microcontroller behind a patient-authenticated on-device menu, publishing telemetry over MQTT to a ThingsBoard cloud dashboard. Relative to the baseline platform, the system improves ambient-temperature sensing accuracy ($\pm\mathbf{0.1}$ °C vs. $\pm\mathbf{0.75}$ °C), replaces the baseline's separate 8-bit microcontroller and Wi-Fi module with a single Wi-Fi-capable ESP32 running MQTT in place of HTTP, and moves from a proprietary paid cloud subscription to an open-source dashboard. A companion Android application concept for multi-patient dashboards and threshold-based alerting is also described. This paper reports the hardware design, firmware architecture, and cloud integration of the prototype; clinical validation against a reference instrument is identified as future work.

Keywords: IoT, Healthcare monitoring, Obesity, ESP32, MQTT, ThingsBoard

1 Introduction

Obesity affects over 650 million adults worldwide and increases the risk of cardiovascular and metabolic complications [2]. Continuous monitoring of vital signs, including blood-oxygen saturation (SpO_2) and heart rate, together with ambient temperature logging, can support early detection of complications, but commercial remote patient monitoring (RPM) devices remain cost-prohibitive for many deployments. This paper presents a low-cost IoT-based monitoring prototype for obese adults that builds on the platform proposed by Gupta et al. [1]. Relative to that baseline, four improvements are targeted:

- higher-accuracy digital temperature sensing (±0.1 °C vs. ±0.75 °C);
- MQTT-based telemetry in place of HTTP polling;
- an open-source ThingsBoard cloud dashboard in place of a proprietary subscription service;
- a rechargeable Li-ion power supply in place of a fixed 12 V DC adapter.

This paper makes the following contributions:

1. A hardware and firmware redesign of the baseline platform using an ESP32 microcontroller, a DS18B20 digital thermometer, and a MAX30102 pulse oximeter, with an on-device patient-login menu (Sect. 3).
2. An MQTT/ThingsBoard cloud-integration layer with a defined topic hierarchy and JSON telemetry schema (Sect. 4).
3. A concept design for a multi-patient Android dashboard application with threshold-based alerting (Sect. 5).

2 Related Work

The baseline platform proposed by Gupta et al. [1] combines an Arduino ATmega328 microcontroller paired with a separate ESP8266 Wi-Fi module, an LM35 analogue temperature sensor (±0.75 °C over its full operating range), a wrist-worn blood-pressure cuff, and HTTP transmission to a proprietary cloud service (iotgecko.com). It established feasibility for obesity-focused wearable monitoring, but four limitations motivate the present work: (i) the 8-bit AVR core has no native wireless connectivity and requires this external ESP8266 radio module as a second chip; (ii) the analogue thermometry chain is limited to ±0.75 °C accuracy by ADC quantisation noise; (iii) the publication does not specify transport-layer security for the Wi-Fi/cloud link; and (iv) the cloud subscription incurs recurring licensing cost around a closed dashboard.

Table 1 Baseline vs. enhanced platform (component-level comparison).

Feature	**Baseline [1]**	**This work**
Microcontroller	Arduino ATmega328 + ESP8266 Wi-Fi module (two chips)	ESP32, single chip (dual-core, integrated Wi-Fi/BLE)
Temperature sensor	LM35 (±0.75 °C)	DS18B20 (±0.1 °C)
Cloud platform	iotgecko.com (proprietary)	ThingsBoard (open-source)
Transport	HTTP	MQTT[1]
Power source	12 V DC adapter	Li-ion battery / 12 V DC adapter
BP sensing	Wrist cuff	Clinical arm cuff[2]

[1] The demonstrated prototype connects to the ThingsBoard demo broker for development; TLS on the broker's secure port is the intended production configuration (Sect. 4).

[2] Hardware slot present on the board design; firmware integration is future work (Sect. 7).

3 System Design

3.1 Hardware Architecture

Figure 1 shows the end-to-end data path. The microcontroller is an ESP32-WROOM-32D development board (dual-core Xtensa LX6, integrated Wi-Fi/BLE), replacing the Arduino ATmega328 and its external wireless module. Ambient temperature is sampled by a DS18B20 waterproof digital probe over a 1-Wire bus, removing the ADC quantisation noise of the baseline's analogue LM35 chain. SpO_2 and heart rate are sampled by a MAX30102 pulse-oximeter module over I^2C, accessed through a MAX30100-compatible pulse-oximeter driver library. A four-button matrix keypad and a 20×4 character I^2C LCD provide local patient interaction (DS18B20 data line on GPIO4; keypad keys 1–4 on GPIO27, GPIO26, GPIO12, and GPIO14; LCD at I^2C address 0x27).

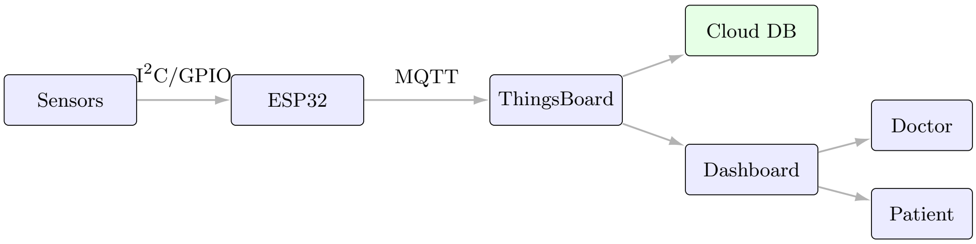


Fig. 1 System architecture: on-board sensors feed the ESP32, which publishes MQTT telemetry to ThingsBoard; the cloud dashboard is consumed by clinicians and patients.

3.2 Firmware Architecture

The firmware implements a menu-driven state machine, shown live on the on-device LCD in Fig. 3. On boot, the patient must enter a 4-digit code on the keypad before any measurement is available; a successful verification unlocks the main menu (Fig. 3a), from which the patient selects SpO_2, heart rate (BPM), temperature, or data transfer. Each measurement runs as a non-blocking 5-second state (Fig. 3b) so that the pulse-oximeter driver (`pox.update()`) and keypad polling continue to execute during the on-screen countdown animation, rather than blocking on a fixed `delay()`, before the result is displayed (Fig. 3c). On "Data Transfer", the device assembles a JSON payload from the most recent readings and publishes it to the cloud (Sect. 4).
Wi-Fi and MQTT broker credentials are configured at startup (Listing 1); the device token and network credentials shown here are anonymised.

Listing 1 Wi-Fi and MQTT configuration (credentials anonymised).

```
// Wi-Fi credentials
const char* ssid = "ANONYMIZED_SSID";
const char* password = "ANONYMIZED_PASSWORD";

// MQTT broker (ThingsBoard server)
const char* mqtt_server = "demo.thingsboard.io";
const int mqtt_port = 1883;
const char* mqtt_client_id = "ESP32_Client";
const char* mqtt_username = "ANONYMIZED_DEVICE_TOKEN"; // ThingsBoard device token
```

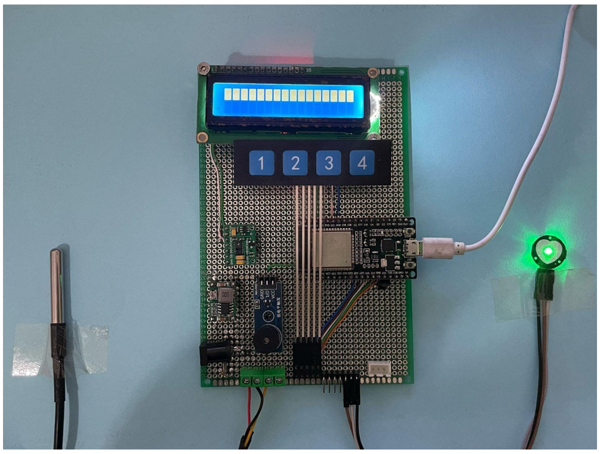
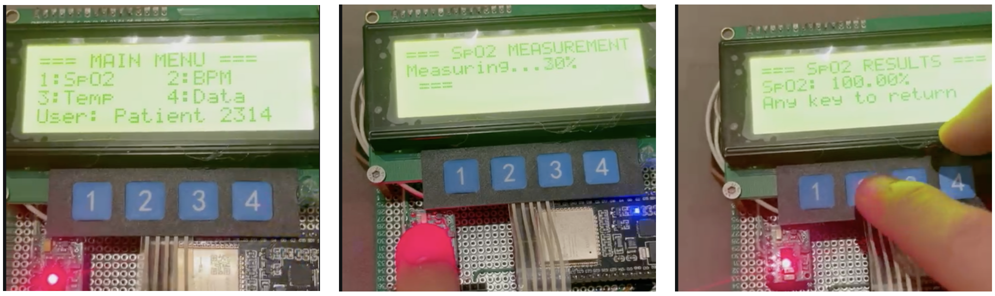


Fig. 2 Assembled prototype: LCD, keypad, ESP32, DS18B20 ambient temperature probe, and MAX30102 pulse-oximeter sensor.



(a) Main menu (b) SpO_2 measuring (c) SpO_2 result

Fig. 3 On-device LCD menu, captured live: (a) main menu after patient login, (b) non-blocking SpO_2 measurement in progress, (c) SpO_2 result screen.

4 Cloud Connectivity and MQTT Implementation

4.1 Telemetry Path

On "Data Transfer", the firmware builds a JSON object from the patient code and the most recent SpO_2, heart-rate, and temperature readings, and publishes it on ThingsBoard's fixed device telemetry topic (Listing 2).

Listing 2 JSON payload construction and MQTT publish (device firmware).

```
String payload = "{";
payload += "\"patient_id\": \"" + patientCode + "\",";
payload += "\"SpO2\": " + String(oldSpO2) + ",";
payload += "\"BPM\": " + String(oldBPM) + ",";
payload += "\"Temperature\": " + String(oldTemp);
payload += "}";

client.publish("v1/devices/me/telemetry", payload.c_str());
```

The device authenticates to the broker using its ThingsBoard access token as the MQTT username, following ThingsBoard's standard device-token authentication scheme [3]. The demonstrated prototype connects to the public demo broker (`demo.thingsboard.io`) on port 1883 for development; the broker's TLS listener on port 8883, with the same token-based authentication, is the intended configuration for a production deployment.

4.2 Why ThingsBoard

ThingsBoard was selected over commercial alternatives (e.g. AWS IoT, Azure IoT Hub) for its open-source licensing, drag-and-drop dashboard builder, and built-in device-provisioning tools, avoiding the recurring per-device subscription fee of the baseline platform's cloud service. Figure 4 shows the resulting dashboard, with live SpO_2, heart-rate, and temperature widgets and a rolling line chart.

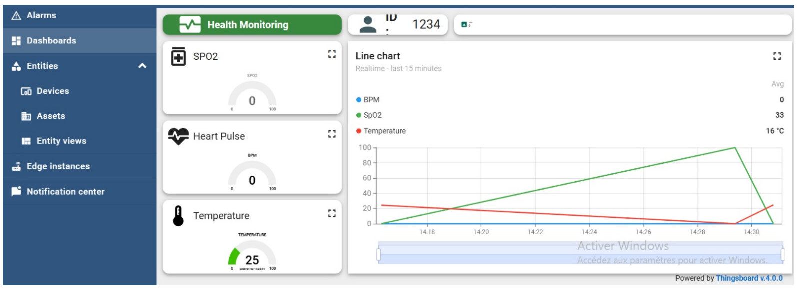


Fig. 4 ThingsBoard patient-monitoring dashboard: live vital-sign widgets and a rolling 15-minute line chart.

4.3 Why MQTT

MQTT's fixed binary header (as small as 2 bytes) is an order of magnitude smaller than HTTP's textual headers, and its publish/subscribe model avoids the round-trip overhead of polling-based protocols [7]. Quality-of-Service level 1 ("at least once") is used for telemetry publishing, giving guaranteed delivery without the duplicate-suppression overhead of level 2.

4.4 Topic Hierarchy

Two MQTT paths are used in the system: the firmware publishes combined telemetry to ThingsBoard's fixed ingestion topic (`v1/devices/me/telemetry`, Listing 2); a second, per-vital topic hierarchy on a public HiveMQ broker is used by the companion Android application (Fig. 5, Sect. 5):

`medmonitor/patient01/spo2`, `medmonitor/patient01/hr`, `medmonitor/patient01/temp`

5 Mobile Application

An Android companion application is under development as the primary patient/clinician interface (Fig. 6). Planned features are: real-time vital-sign cards,

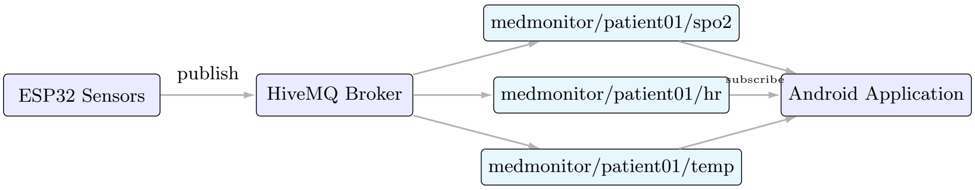


Fig. 5 MQTT topic hierarchy for the Android companion application: the ESP32 publishes per-vital telemetry to three topics on a HiveMQ broker, each independently subscribed to by the mobile app. This path is separate from the direct ESP32-to-ThingsBoard MQTT link used by the web dashboard (Fig. 1).

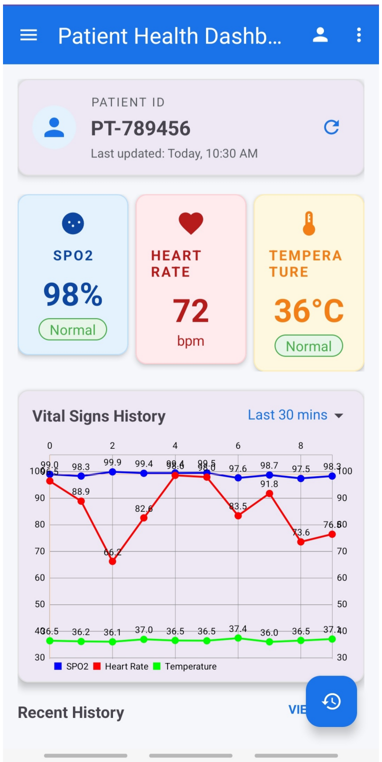


Fig. 6 Android application interface: vital-sign cards and a 30-minute history chart, captured live during a test session.

a rolling historical chart, threshold-based colour-coded alerts ($SpO_2 < 90\,\%$, heart rate > 100 bpm, ambient temperature $> 38\,°C$), and multi-patient profile switching. The application subscribes to the per-vital HiveMQ topics of Fig. 5; during development, the broker's web console confirmed three active topic subscriptions corresponding to the SpO_2, heart-rate, and temperature topics.

6 Discussion

Comparative Analysis. Relative to the baseline of Gupta et al. [1] (Table 1), the prototype delivers four substantive advances: (i) the baseline's two-chip 8-bit AVR plus ESP8266 Wi-Fi module combination is replaced with a single 32-bit dual-core ESP32 featuring integrated Wi-Fi/BLE, eliminating the separate radio module and enabling on-device signal processing; (ii) analogue LM35 thermometry ($\pm 0.75\,^{\circ}\mathrm{C}$) is upgraded to digital DS18B20 sensing ($\pm 0.1\,^{\circ}\mathrm{C}$), removing ADC quantisation noise as a measurement error source; (iii) HTTP polling to a proprietary cloud service is replaced with MQTT publishing to an open-source ThingsBoard dashboard, eliminating recurring subscription costs; and (iv) a patient-authenticated on-device login menu introduces access control absent from the baseline design.

Current Limitations and Mitigation. While the prototype demonstrates technical feasibility, four limitations warrant acknowledgement and guide future work:

Clinical validation. Measurement accuracy has not been statistically validated against a reference instrument; the system has been exercised only on a single benchtop prototype. Mitigation: controlled trials against clinical-grade vital-signs monitors (Sect. 7).

Transport security. The MQTT link currently uses the broker's default port for development; production deployment will employ TLS on port 8883 (Sect. 7).

Blood-pressure integration. Hardware for BP sensing (clinical arm cuff) is present on the board design but firmware integration remains outstanding (Sect. 7).

Usability evaluation. The Android companion app has been exercised with live sensor data but has not undergone formal usability testing with patients or clinicians.

These limitations do not undermine the architectural and security improvements demonstrated; rather, they define a clear path toward clinical-grade deployment.

7 Future Perspectives

Future work centers on three pillars: (i) Clinical validation through multi-center trials against reference vital-signs monitors; (ii) Production hardening via TLS-secured MQTT (port 8883), custom PCB design for reduced cost and power consumption, and integration of blood-pressure (HX711), ECG (AD8232), non-invasive glucose, accelerometer-based fall detection, and ambient humidity sensors; and (iii) Intelligence augmentation through predictive machine-learning models trained on longitudinal data to forecast patient-specific health-risk trajectories, extending beyond simple threshold-based alerts.

8 Conclusion

This paper presented an IoT-based healthcare monitoring prototype for obese adults that improves on the platform of Gupta et al. [1] through: (i) MQTT-based telemetry in place of HTTP polling; (ii) high-accuracy digital ambient-temperature sensing (DS18B20, $\pm 0.1\,^{\circ}\mathrm{C}$); (iii) a patient-authenticated on-device interface feeding a JSON telemetry pipeline to an open-source cloud dashboard; and (iv) a concept design for a multi-patient mobile dashboard with threshold-based alerting. Clinical validation of measurement accuracy, TLS-secured production deployment, and blood-pressure sensing integration are identified as near-term future work.

Declarations

- Funding: Not applicable.
- Conflict of interest/Competing interests: The authors declare no competing interests.
- Ethics approval and consent to participate: Not applicable; the study did not involve experiments on live vertebrates, higher invertebrates, human participants, or human samples.
- Consent for publication: The individuals appearing in device photographs are members of the author team and have consented to the use of these images.
- Data availability: Not applicable.
- Materials availability: Not applicable.
- Code availability: The ESP32 firmware source is available from the corresponding author upon reasonable request.
- Author contribution: All authors contributed to the system design, implementation, and manuscript preparation.

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